Field2VC: Automating the Generation of ASPICE-Compliant Verification Criteria from Field Failures

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***Abstract*—In Automotive SPICE, every SYS.2 system requirement must be verifiable through explicit verification criteria (VCs); for safety-relevant ECUs, ISO 26262 [1] reinforces this obligation. Yet VCs are systematically incomplete: engineers rarely foresee failure modes they have not previously encountered. We present Field2VC, which grounds LLM-based VC generation in two complementary failure sources—developer-reported defects on GitHub, which expose code-level boundary conditions absent from specifications, and consumer complaints in the NHTSA ODI database, which expose vehicle-level failures engineers do not anticipate. Field2VC filters each report into a structured failure representation, retrieves the failures most relevant to a requirement, and generates specific, testable VCs. In a four-expert evaluation of 489 generated VCs over a 350-requirement industrial benchmark, failure grounding raises expert acceptance from a 45.3 % no-context baseline to 65.5 %–67.2 % (odds ratio 2.3–2.5×; Holm-corrected $p < 0.001$). An adjudication-sensitivity analysis shows this gain is uneven: on the adjudication-independent agreed subset, the GitHub channel is accepted on every item (37/37 versus 93 % for the baseline) while the NHTSA channel falls to 51 %—the GitHub gain is robust, the NHTSA gain partly adjudication-dependent. The broader lesson is that a failure source's usefulness depends on vocabulary alignment: each source is effective on the requirement subdomain whose language its filtered failures match—developer issues for the communication and diagnostics stack, regulator complaints for body-control. We release the benchmark, all generated VCs, and per-rater labels.**

***Index Terms*—verification criteria, automotive requirements, ASPICE SYS.2, issue mining, LLM, failure knowledge, ISO 26262, NHTSA**

I. Introduction

In automotive software engineering, the cost of repairing a defect rises sharply the later it is found; classic, if debated, estimates put a post-delivery fix at up to 100 times the cost of a requirements-phase fix [2] (Fig. 1). The obligation, however, does not rest on any particular cost multiplier. ASPICE SYS.2 strictly requires every system requirement to be *verifiable*: a *Verification Criterion* (VC) must coexist with the requirement text [3]. For safety-relevant ECUs this obligation is reinforced by ISO 26262, whose functional-safety lifecycle requires each requirement to be verified before integration [1].

Writing thorough VCs, however, fails in a specific and repeatable way. A requirement specifies intended behaviour under nominal conditions; a complete VC must additionally probe boundary conditions and fault modes. Reviewers reason forward from the requirement text, so the criteria they write can only confirm behaviours the specification already anticipates, while field failures, almost by definition, occur where it did not. The blind spot is structural rather than a matter of effort: a VC that omits an unforeseen edge case passes internal review precisely because no reviewer holds the evidence that would expose it (Fig. 1).

Closing this gap requires failure evidence from outside the project, matched to the abstraction level at which a SYS.2 requirement is verified—the *system* level, not the software unit. Defect trackers are a natural candidate, mined by recent work such as CrUISE-AC [4] to enrich agile acceptance criteria: developer-reported defects describe faults as engineers observe them—isolated to a module, reproducible, often with a root cause. But failures emerging from cross-subsystem interactions and months of customer usage rarely surface in any repository, because no developer observes the deployed fleet. Regulator-collected complaints capture exactly this stratum: NHTSA's ODI database records faults as the vehicle exhibits them, at fleet scale, under conditions no test plan enumerates [5], [6]. The two corpora are thus not interchangeable; our evaluation confirms the split empirically—complaint-grounded VCs are accepted more often for body and chassis requirements, defect-grounded VCs for communication and diagnostics (§V-B). We use NHTSA as a public, reproducible proxy for the warranty and field-return data an OEM holds internally but cannot release; the pipeline is agnostic to the vehicle-level channel's source. Manual inspection of 200 sampled requirement–evidence pairs confirms this mapping stage at **73.5 %** precision (§IV).

This motivates our central question: given a system requirement, can generation grounded in cross-level field-failure evidence produce verification criteria that practising engineers would accept into a SYS.2 baseline? We treat expert acceptance as the measurable proxy for usefulness, since no oracle for "complete" VC coverage exists.

This paper presents **Field2VC**, an automated VC generation mechanism grounded in real failure evidence. Field2VC extracts defect context from two channels—GitHub (code-level failures) and NHTSA (vehicle-level failures)— filters domain-specific noise from the raw records, and uses the retained context as a semantic constraint so a large language model (LLM) generates verifiable VCs covering historical failure scenarios.

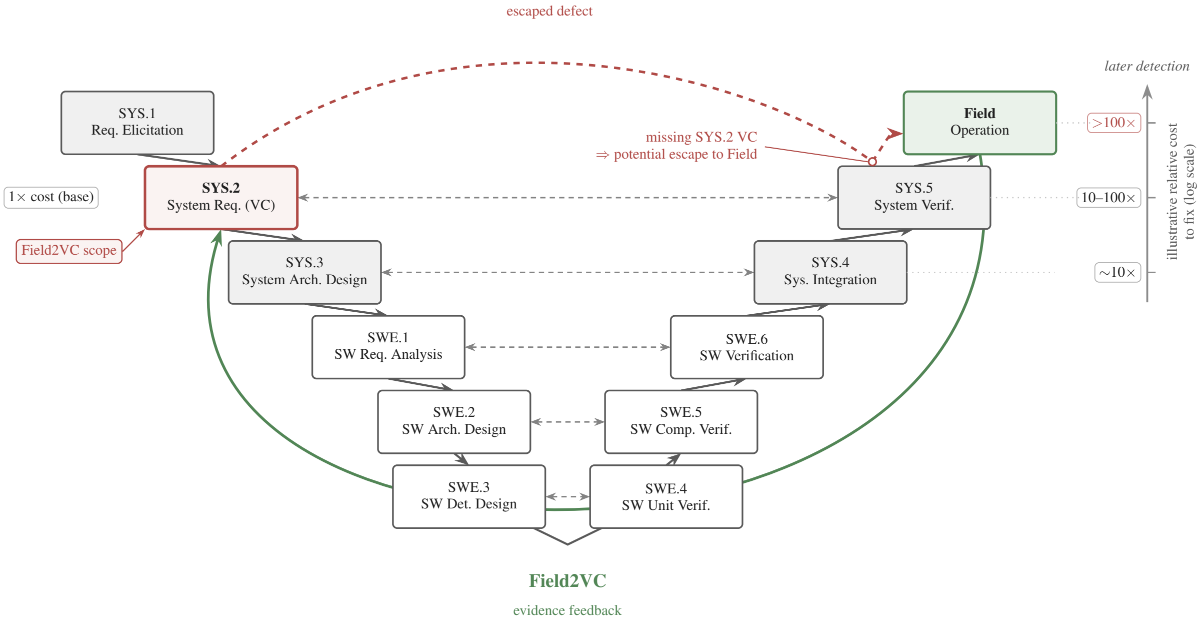

Fig. 1. ASPICE V-model cost escalation. A missing VC that passes SYS.2 review surfaces only at system-level testing or as a field complaint, where re-verification is far costlier. Field2VC shifts detection left by grounding VC generation in field-failure evidence *at* SYS.2 (the *Field2VC scope* annotation); the green arc is the evidence feedback loop from field operation back to SYS.2. SWE-layer processes (SWE.1–SWE.6) are out of scope.

a) Contributions.:

- **An empirical finding**: a four-expert evaluation of 489 generated VCs (307 unique industrial requirements) shows that field-failure grounding raises expert acceptance over a no-context baseline, and yields a design rule—a failure source is useful only where its filtered vocabulary aligns with the target requirement subdomain (§V).
- **An automated, evidence-grounded pipeline** combining developer-reported defects (GitHub) and regulator-collected complaints (NHTSA) at distinct abstraction levels, with an abstention mechanism that declines when retrieved evidence is insufficient (§III).
- **An industrial benchmark**: 350 de-identified SYS.2 requirements (307 evaluated) with expert-annotated golden VCs, all generated VCs, per-rater labels, and analysis scripts (Data Availability).

Our results extend work on issue-tracker-driven requirements enrichment [4] in two directions that prior work does not cover: the evidence is drawn from two channels at *distinct abstraction levels*, and the target artefact is an ASPICE-mandated verification criterion for a safety-relevant ECU rather than an agile acceptance criterion.

b) Data Availability.: The de-identified replication package is available *now* for review at https://anonymous.4open.science/r/apsec2026-artifact-C562/, containing: (1) the System Requirements Specification (SYRS) corpus (all 350 benchmark items) with expert-annotated golden VCs; (2) all 489 generated VCs with per-VC rater labels, majority-vote labels, and the coordinating expert's adjudication record; (3) the three-criterion evaluation guidelines; (4) pipeline scripts for Stages 2–5 and all prompt templates; and (5) statistical analysis scripts. Proprietary signal names are masked by stable placeholders; the redaction script is part of the package.

II. Related Work

A. Automated Requirements Completion within Specifications

Requirements completeness research falls into two streams. *Refinement* approaches improve individual requirements through semantic enrichment without adding new external data, using techniques such as NL-level templating and domain-knowledge-driven pattern systems, e.g., the EARS controlled requirements syntax [7]. *Derivation* approaches generate new requirements by exploiting relationships within or between existing requirement items.

LLM-based derivation has advanced rapidly [8]. F2SRD [9] retrieves the applicable OWASP ASVS verification requirements for each functional specification and guides GPT-4 to derive security requirements grounded in them. Such approaches draw their completion signal from within-specification logic or applicable standards, not from empirical failure evidence; our work is complementary: we exploit *external* failure repositories to surface conditions that no single specification can reveal.

Closest in spirit is TVR [10], which applies retrieval-augmented generation (RAG) to validate and recover missing traceability links between stakeholder and system requirements in industrial automotive projects. TVR targets *traceability* (linking existing requirements) rather than *VC completion* (generating new complementary criteria); its retrieval corpus is the specification itself, whereas ours is external failure evidence. These two directions show that RAG-based requirements engineering is being applied across multiple automotive sub-tasks [11], [12].

B. Enriching Requirements from External Repositories

Mining bug reports and issue trackers for RE has a long history [13], [14]. Do et al. [15] extract requirements from similar products via doc2vec and BIRCH clustering; Almonte et al. [16] prompt LLMs to derive quality-attribute NFRs from functional specifications; Koscinski et al. [17] use RelGAN to transform functional information into security requirements. These mine requirements-like data from whatever corpora are available; we instead mine two repositories from *different domains*—a defect tracker and a regulator-operated complaint database—and show the knowledge source strongly predicts VC quality.

CrUISE-AC [4] is the leading crowd-data-driven system over external repositories, mining public issue trackers (seven e-commerce systems, >165 k raw issue records, plus two content-management systems) for issues matching Gherkin acceptance criteria, achieving 82 % approval in e-commerce under a four-expert ($\geq$3-of-4) majority vote with Gwet's $\text{AC1} = 0.74$. Wang et al. [18] likewise generate acceptance criteria from multi-modal requirements with LLMs, but target user-facing rather than safety-critical requirements.

Field2VC differs from CrUISE-AC in three respects. First, our domain is the automotive system-requirements level (ASPICE SYS.2) for safety-relevant ECUs. Second, we evaluate against a three-criterion acceptance protocol (Completeness ≥ 3, Novelty = Y, Usefulness = Y) rather than a binary judgement, making direct percentage comparisons misleading. Third, our ablation contrasts LLM classification and embedding matching as grounding mechanisms, primarily exposing the effect of the filtering-enabled abstention decision rather than the matching method alone (§V-C).

C. LLM-based Requirements Engineering

Recent work applies LLMs to diverse software engineering tasks [19], [20]: MARE [21] proposes a multi-agent collaboration framework for requirements elicitation. Like the others, it targets neither VC completion nor the effect of grounding in external failure evidence—the key contribution of Field2VC. In the automotive-safety domain specifically, LLMs have been applied to code generation and verification [22], [23], functional-safety and security design [24], and RAG-based derivation of safety requirements from standards [25]; these operate on code or upstream safety requirements rather than completing verification criteria for existing SYS.2 specifications.

III. Field2VC: Two-Channel VC Completion Pipeline

Field2VC is a five-stage pipeline that transforms raw failure records from two complementary external channels into candidate missing Verification Criteria (VCs) for automotive ECU SYRS items, each traceable to one or more source records (Fig. 2). We illustrate it throughout with a real Unified Diagnostic Services (UDS, ISO 14229) Security Access requirement (SYRS.1018_A) and its GitHub field failure (`python-udsoncan#51`)—a Channel A trace that surfaces protocol-transport boundary conditions absent from specification. Table I complements it with a Channel B (NHTSA) trace, where complaint evidence surfaces a real-world electromechanical scenario the golden VC did not anticipate.

TABLE I
Worked trace: an accepted Channel B (NHTSA) VC traced through all five stages. Failure text is verbatim from the public source; SYRS and golden VC are paraphrased from the de-identified corpus.

Channel B: NHTSA, SYRS.0177 (low-beam robustness under AC flicker). *SYRS:* do *not* switch off low-beam headlights when exposed to synthetic AC-flicker light sources if ambient brightness remains < 1000 lux for more than 4 s. The golden VC tests the *nominal AC-flicker path only*.
Matched failure: ODI #11557628 (sim. 0.60; 2022 Toyota Camry): "low beam headlights flashing when high beams are on ... also flash on ignition."
Triple: $\langle$high-beam activation or ignition key-on | low-beam LEDs remain steadily illuminated | bilateral intermittent flashing (right front more severe); state-stuck fault$\rangle$.
Generated VC: fault-injection in an AC-flicker environment ($<$1000 lux sustained $>$4 s): activate high beams and separately cycle the ignition, verifying that low beams neither flash nor extinguish in either case—electromechanical interaction scenarios absent from the golden VC. **Accepted** (E3: C=3; E4: C=3; N = Y, U = Y; unanimous).

A. Stage 1: Issue Collection

We define two channels providing complementary failure evidence:

Channel A: GitHub Issues. We index issues from four open-source repositories covering ECU-adjacent stacks: `pylessard/python-udsoncan` (UDS diagnostics), `python-can-isotp` (ISO-TP transport), `hardbyte/python-can` (CAN bus), and `ecubus/EcuBus-Pro` (ECU test toolchain). We retrieved **696 issues** via the GitHub REST API,[1] prioritising closed issues labelled {bug, defect, regression, safety} and supplementing with unlabelled closed issues for coverage. The running example fetches `python-udsoncan#51`: *"SecurityAccess does not handle already-unlocked server."*

Channel B: NHTSA Complaints. We query the NHTSA ODI consumer-complaint database [26] using component keywords matching automotive ECU subsystems. We retrieved **364 complaint records**, each containing a free-text consumer narrative plus structured vehicle and component metadata. Unlike curated recall notices, complaint narratives are raw, colloquial, and noisy field reports, which is why the Stage-2 filter matters most for this channel.

[1] https://docs.github.com/en/rest/issues

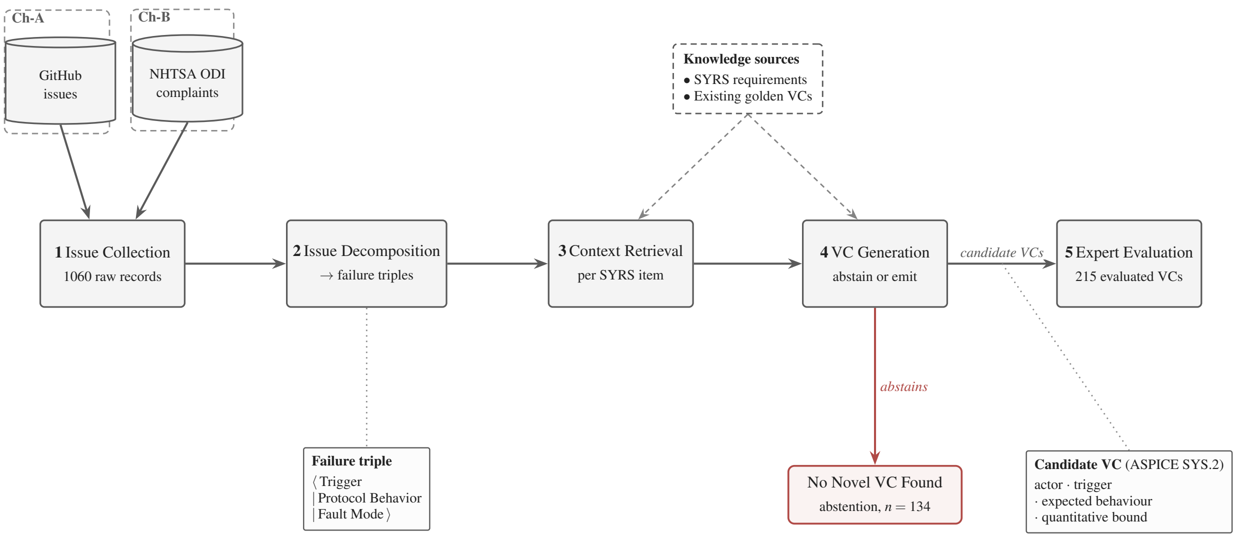

Fig. 2. Field2VC five-stage pipeline. Failure records from GitHub (Ch-A) and NHTSA (Ch-B) are filtered into ⟨Trigger | Protocol Behavior | Fault Mode⟩ triples (Stage 2, Qwen-Plus), embedding-matched to each SYRS item via cosine similarity (Stage 3, `text-embedding-v3`), and used to prompt VC generation (Stage 4, Qwen-Max). The generator is never forced to produce a VC: it may decline by emitting the fixed marker `NO_NOVEL_VC_FOUND` when the evidence reveals no genuinely missing condition, the pipeline's only relevance gate. Four domain experts review the surviving VCs under the three-criterion acceptance protocol (Stage 5). The diagram traces the 215 channel VCs (Ch-A + Ch-B); the full study adds 274 no-context baseline VCs for 489 total (Table II).

B. Stage 2: Issue Decomposition

We use an LLM (Qwen-Plus) to filter each raw record into a structured triple:

$$\langle \textit{Trigger} \mid \textit{Protocol Behavior} \mid \textit{Fault Mode} \rangle$$

Trigger is the activating condition, *Protocol Behavior* the expected ECU action per the relevant standard, and *Fault Mode* the observed deviation; filtering reduces vocabulary mismatch and suppresses implementation-layer detail irrelevant to SYS.2. The core instruction is *"Extract the technical essence from this report"*, with a mandatory escape route: records not about automotive system behaviour (e.g., Python packaging, documentation, consumer venting) return all three fields as `null`[2] and are dropped—the point where off-topic noise leaves the pipeline. The running example, issue #51, reduces to ⟨`0x27` requestSeed while already unlocked | client detects all-zero seed and skips sendKey | key derived from zero seed → Negative Response Code (NRC) `0x24` rejection⟩.

C. Stage 3: Context Retrieval

For each SYRS item s and filtered triple t_i we compute cosine similarity [27], [28] using `text-embedding-v3` (Alibaba DashScope; snapshot April–May 2026), retrieving the top-$K = 5$ triples as the *failure context* $\mathcal{F}(s) = \{t_1, \ldots, t_5\}$; K=5 is a pragmatic trade-off between computational cost and retrieval diversity. We impose *no* hard similarity cut-off: every item proceeds with its top-5 context and the relevance decision is delegated to generation [29], where the model abstains (`NO_NOVEL_VC_FOUND`) if the evidence reveals no genuinely missing condition (§III-D). This mirrors, post-retrieval, the pre-retrieval LLM relevance classifier of CrUISE-AC [4]—a choice our ablation examines directly (RQ3, §V-C). For SYRS.1018_A, issue #51 ranks second at cosine 0.69 and enters the top-5 context. Embedding matching has two advantages over LLM binary classification: it scales to thousands of pairs without per-pair LLM calls, and it yields a continuous, auditable relevance score released with every match.

[2]All Stage-2 and Stage-4 prompts are released verbatim in the replication package (`prompts/`).

D. Stage 4: VC Generation

We prompt Qwen-Max with three inputs: the SYRS item, its existing golden VCs (as *do-not-repeat* context operationalising "missing"), and the failure context $\mathcal{F}(s)$. The prompt requires VCs that (a) genuinely test the requirement (not the protocol library the failure was reported against), (b) are not redundant with the golden VCs, and (c) are inspired by a real failure pattern in $\mathcal{F}(s)$ and testable at ECU system-integration level. It closes with explicit permission to decline—*"if no genuinely useful missing VC can be identified, output NO_NOVEL_VC_FOUND"* (the *sentinel*). Declining is expected, not an error: rather than hallucinate a plausible criterion [30], the pipeline abstains when the evidence is irrelevant. For the running example, the model turns issue #51's triple into a negative test of the already-unlocked path (`0x27` at L1 unlocked, send `0x31 01 10 08`, expect no NRC `0x33` securityAccessDenied)—a scenario neither the SYRS nor its golden VC exercises.

E. Stage 5: Expert Evaluation

Each generated VC is independently scored by at least two of the three external raters (E2, E3, E4) on three dimensions under a **three-criterion acceptance protocol**:

- **C (Completeness)**: Score 1–5. Does the SYRS item omit this VC? ($C \geq 3$ is the acceptance threshold.)
- **N (Novelty)**: Y/N. Is this VC absent from the existing SYRS item and its associated VCs?
- **U (Usefulness)**: Y/N. Is this VC actionable for downstream test design and software qualification?

A VC is accepted if and only if $C \geq 3$ AND $N = \mathrm{Y}$ AND $U = \mathrm{Y}$. The label is the majority vote (MV) among the raters who scored each VC; E1 (adjudicator) resolves disagreements on the N and U dimensions. The running-example VC was accepted (C=4, N=Y, U=Y) after E1 adjudicated a novelty split between E2 and E3.

F. Ablation Variants

Two ablation conditions isolate the contributions of filtering and embedding matching. **Abl-NF (No Filter)** skips Stage 2, embedding the raw issue title + body (first 500 characters) and matching to SYRS by cosine similarity—testing whether raw-text embedding suffices. **Abl-LM (LLM-Match)** replaces Stage 3 similarity with an LLM binary relevance classifier (Qwen-Plus), replicating CrUISE-AC's matching [4] to test LLM judgement against embedding similarity.

IV. Experiment Design

A. Research Questions

- **RQ1 (Effectiveness):** Does failure-grounded prompting generate VCs with significantly higher expert acceptance than a no-context baseline?
- **RQ2 (Channel–Domain Fit):** The two channels are mapped to disjoint requirement subdomains by construction. Within each subdomain (body-control/chassis, communication stack/state management, diagnostics), how effective is the assigned channel relative to the no-context baseline, and how do the two channels divide labour across subdomains?
- **RQ3 (Ablation):** What is the contribution of (a) failure-triple filtering and (b) embedding-based matching to VC quality?
- **RQ4 (LLM Sensitivity):** How stable are the channel-level results across different LLMs from multiple model families?

B. Dataset

a) SYRS corpus.: We apply the pipeline to 350 unique automotive ECU SYRS items drawn from a proprietary corpus of 675: the 350 are every item falling in the two channels' target subdomains, with items in other domains (e.g., powertrain, ADAS, gateway) out of scope (§VI). The channel mapping is disjoint by construction: Ch-A = {communication stack (UDS, CAN), diagnostics} (208 items) and Ch-B = {body control (steering-column lock, parking brake, steering-column control module), IO/HMI, functional safety, system-state management} (142 items). This canonical mapping governs every per-category breakdown (Table IV). All items follow the ASPICE SYS.2 format (explicit actor, trigger, quantitative boundary, expected behaviour) and carry human-annotated golden VCs.

b) Generated VCs.: Across both channels and the no-context baseline, 489 VCs entered expert evaluation (Table II):

TABLE II
DATASET AND PER-CONDITION VC FLOW. CH-A AND CH-B COVER DISJOINT SYRS SUBSETS ($208 + 142 = 350$) AND ABSTAIN VIA THE NO_NOVEL_VC_FOUND SENTINEL; "CAND." COUNTS CANDIDATE VCS AND THE CAND. → EVAL. GAP REMOVES DUPLICATE/MALFORMED CANDIDATES. SOURCES: 696 GITHUB ISSUES, 364 NHTSA COMPLAINTS.

Condition	Items	Abst.	Cand.	Eval.
A (GitHub)	208	121	88	87
B (NHTSA)	142	13	131	128
Baseline (all SYRS)	350	73	277	274

Evaluation totals $87 + 128 + 274 = 489$ VCs spanning 307 unique SYRS; 43 items yielded no evaluated VC. Ch-A and Ch-B share no SYRS; the baseline spans all 350 items (overlapping both channels).

C. Evaluation Metrics

a) Accept rate.: Fraction of VCs whose majority-vote label is Accept (three-criterion protocol: $C \geq 3$ AND $N = \mathrm{Y}$ AND $U = \mathrm{Y}$).

b) Wilson 95 % CI.: Confidence interval for the accept rate using the Wilson score method [31].

c) Fisher's exact test + Holm–Bonferroni correction.: Each channel is tested against the baseline with Fisher's exact test [32]; p-values are corrected with Holm–Bonferroni [33] across three simultaneous comparisons (Ch-A, Ch-B, and A+B combined vs. baseline).

d) Cohen's h.: Effect size for proportion comparisons [34]: $h \geq 0.2$ small; $h \geq 0.5$ medium; $h \geq 0.8$ large.

e) Generation-rate proxy.: The *generation rate* is the fraction of SYRS items for which the model generates a VC rather than abstaining (`NO_NOVEL_VC_FOUND`). Under active filtering a high rate indicates genuinely new failure angles, while near-universal generation (>93 %) signals the model is ignoring the abstention instruction. We use it as a lightweight quality proxy for the RQ4 sensitivity runs, where full human evaluation is unavailable.

f) Abstention rates.: Ch-A abstains on 121 of 208 SYRS items (58.2 %), Ch-B on 13 of 142 (9.2 %). The gap reflects vocabulary alignment: once filtered, NHTSA fault modes map tightly to body-control and chassis items, whereas GitHub vocabulary is more varied relative to communication-stack and diagnostics items—the pipeline generates selectively, not indiscriminately.

g) Retrieval precision.: To gauge upstream retrieval quality, a single author manually evaluated a stratified random sample of 200 requirement–evidence pairs (100 per channel, seed = 42, similarity 0.45–0.78); the absence of a second annotator here is a construct-validity limitation. A pair is labelled *relevant* if the retrieved evidence describes a failure mode or boundary condition observable in the ECU under test (not merely a topically adjacent software defect or host-toolchain issue). The pipeline achieves an overall matching precision of **73.5 %** (Ch-A GitHub: **61.0 %**; Ch-B NHTSA:

86.0 %), confirming that the evidence reaching the generator is failure-relevant.

D. Inter-Annotator Agreement

a) Rater profiles.: Four raters span the three ASPICE Verification & Validation (V&V) stakeholder roles (process ownership, academic methodology, hands-on practice), guarding against single-perspective bias: **E1** (Embedded Systems Project Manager, 10 yr) adjudicates split votes under defined criteria; **E2** (ASPICE assessor and functional-safety consultant) contributes SYS.2 process authority; **E3** (PhD candidate in automotive software testing) provides methodology-grounded judgement; **E4** (senior functional-test engineer, OEM/Tier-1) anchors physical V&V executability. This industry–academia panel replicates CrUISE-AC's design [4].

b) Main evaluation (RQ1/RQ2).: E2, E3, and E4 independently scored the 489 VCs (each receiving two independent external ratings); E1 resolved split decisions under the three-criterion protocol, and the label is the majority vote among scoring raters. We use Gwet's AC1 [35], [36] rather than Cohen's κ to avoid the kappa paradox [37] under high Accept-class prevalence, reported on Completeness ($C \geq 3$).

c) Ablation evaluation (RQ3).: The 50-item subset comprises the first 50 Ch-A items in corpus order, all Communication-Stack requirements; this homogeneity strengthens internal comparability (category cannot confound the contrast) but restricts RQ3's external scope to that category (§VI). All four raters (E1–E4) score the subset; the label is the majority vote of E2–E4 (≥2/3) with E1 as adjudicator (matching the main protocol), and all four scores feed the pairwise AC1 analysis.

E. Baselines

a) No-context baseline.: Qwen-Max prompted with SYRS text only (no failure context). This represents the zero-shot VC generation scenario and is the primary comparison point for RQ1.

b) CrUISE-AC [4].: The closest published system, targeting acceptance criteria for user stories from crowd-reported bugs. Accept rates and IAA values are compared in §V-A to contextualise our results.

F. Computational Cost

The core pipeline (Qwen-Plus filtering, Qwen-Max generation on both channels, and the ablations) cost approximately **US$3** in API fees; the seven-model RQ4 sweep added roughly US$12 (dominated by reasoning-heavy models' long completions), for a total on the order of US$15. Embedding matching was negligible (< US$0.05) and the Domain-2 pilot added about US$0.50. Figures are approximate, depending on time-varying provider pricing.

V. Results and Analysis

A. RQ1: Effectiveness of Failure-Grounded Prompting

Table III summarises acceptance for all conditions; both channels significantly outperform the baseline after Holm–Bonferroni correction. Channel A (GitHub) reaches 65.5 % (57/87; OR = 2.30, Cohen's $h = 0.41$, Holm-$p = 0.0007$) and Channel B (NHTSA) 67.2 % (86/128; OR = 2.48, $h = 0.45$, Holm-$p < 0.001$); combined, A+B reaches 66.5 % (143/215; $p < 0.001$, OR = 2.40, $h = 0.43$), confirming both channels contribute.

TABLE III
RQ1: Accept Rates vs. Baseline (three-criterion protocol; majority vote). 95 % CI is the Wilson score interval (percentage points).

Condition	VCs	Acc.	Rate	95 % CI	Holm-p
Baseline (no-context)	274	124	45.3 %	[39.5, 51.2]	—
Channel A (GitHub)	87	57	65.5 %	[55.1, 74.7]	0.0007***
Channel B (NHTSA)	128	86	67.2 %	[58.7, 74.7]	<0.001***
A+B combined	215	143	66.5 %	[60.0, 72.5]	<0.001***

*** $p < 0.001$ (Holm–Bonferroni corrected).

a) IAA.: Pairwise Gwet's AC1 on Completeness ($C \geq 3$) is 0.84 (E2–E3), 0.65 (E2–E4), and 0.69 (E3–E4), averaging 0.73; the two lower pairs both involve E4's physical-executability perspective vs. E2/E3's requirements/process perspectives (a discipline-level finding, §VI). The mean is comparable to CrUISE-AC's AC1 = 0.74 [4], under our strictly stronger three-criterion acceptance predicate.

b) Per-item accepted-VC yield.: Conditional accept rates compare *generated* VCs and are not commensurable across conditions with different abstention rates; the *per-item yield* (accepted VCs over all assigned items) is a common-denominator alternative. Generation rate, accept rate, and yield are 79.1 %/45.3 %/0.35 for the baseline (350 items), 41.8 %/65.5 %/0.27 for Ch-A (208 items), and 90.8 %/67.2 %/0.61 for Ch-B (142 items). Ch-A's yield falls below the baseline—a *precision channel* best combined with the baseline where broad coverage is needed—whereas Ch-B's exceeds it, covering most assigned items at a higher accept rate.

B. RQ2: Channel–Domain Fit

Because the two channels are mapped to disjoint requirement subdomains, RQ2 concerns where each channel fits rather than which channel performs better in a head-to-head comparison. Within its assigned domain, each channel yields a category-dependent uplift over the no-context baseline averaging +20 percentage points (pp): +20.7 pp for Channel A and +21.3 pp for Channel B (category-matched,[3] from Table IV), each backed by a small-to-medium effect size (Channel A $h = 0.41$; Channel B $h = 0.45$). The grounding mechanism differs by source: once Stage-2 filtering removes their

[3] *Pooled* deltas (the +20/+22 pp in RQ1) compare each channel against the full baseline; *category-matched* deltas (RQ2) compare against the baseline restricted to that channel's categories.

colloquial noise, NHTSA complaints' component-tagged fault modes align closely with SYRS vocabulary, whereas GitHub issues offer higher volume (696 vs. 364 records) at the cost of greater terminological variance.

a) Scope-stratified acceptance.: Table IV breaks acceptance down by ECU category. Body-control and chassis items (Ch-B) show consistently high acceptance—IO/HMI 74.2 % (+28 pp), Functional Safety 71.9 %, Body Control 68.3 %—as filtered NHTSA narratives map tightly to sensor-level and state-machine specifications; communication-stack items (Ch-A) improve more modestly (61.3 % vs. 56.4 %, +4.9 pp), reflecting greater GitHub-issue vocabulary variation for protocol-specific SYRS items.

TABLE IV
SCOPE-STRATIFIED ACCEPTANCE BY ECU CATEGORY (THREE-CRITERION PROTOCOL, MAJORITY VOTE). DCM/DEM = DIAGNOSTIC COMMUNICATION/EVENT MANAGER. "—" = NO EVALUATED BASELINE VC FELL IN THIS CATEGORY.

Category	Ch.	Pipeline		Baseline	
		Acc/N	Rate	Acc/N	Rate
Body-Control & Chassis (Ch-B)					
IO / HMI	B	23/31	74.2 %	18/39	46.2 %
Functional Safety	B	23/32	71.9 %	—	—
Body Control	B	28/41	68.3 %	21/42	50.0 %
Comm. Stack & State Mgmt (Ch-A/B)					
Communication Stack	A	19/31	61.3 %	31/55	56.4 %
System State Mgmt	B	12/24	50.0 %	11/28	39.3 %
Diagnostics (Ch-A)					
Diagnostics (DCM)	A	29/44	65.9 %	25/64	39.1 %
Diagnostics (DEM)	A	4/6	66.7 %	14/33	42.4 %
Diagnostic Services	A	5/6	83.3 %	4/13	30.8 %

Table I traces an accepted Channel B VC end-to-end; with the Channel A running example of §III it illustrates the two channels' complementary discovery roles.

C. RQ3: Ablation

Table V reports human acceptance rates on the 50-item Ch-A evaluation subset under the three ablation conditions.

TABLE V
RQ3: HUMAN ACCEPTANCE RATE ON 50-ITEM CH-A SUBSET (4-RATER MAJORITY VOTE, $C \geq 3$ AND N=Y AND U=Y).

Condition	Accepted	Rate
Full Field2VC (FL, filter + embed)	29/50	58.0 %
Abl-LM (LLM-Match*)	29/50	58.0 %
Abl-NF (No Filter, embed)	41/50	82.0 %

*Replicates CrUISE-AC matching strategy [4]: LLM binary relevance classification, no triple filtering.

a) Embedding matching vs. LLM binary classification.: On the same 50-item subset, the clean matching-method comparison is between the two *unfiltered* ablations: Abl-NF (raw-text embedding) achieves 82.0 % versus 58.0 % for Abl-LM (LLM binary classifier, equivalent to CrUISE-AC's strategy [4]), a 24 pp gap attributable to the retrieval mechanism since filtering is absent in both. The full pipeline (FL, filter + embed) matches Abl-LM exactly at 58.0 %: with filtering active, the accept rate is set by the Stage-2 abstention decision, not the matching step.

b) Why filtering is necessary at scale.: We do not dispute the local inversion: on these 50 items, removing the filter (Abl-NF, 82.0 %) scores higher than the full pipeline (58.0 %). That cost is invisible at this scale and surfaces only at corpus scale, where Abl-NF's generation-rate proxy reaches 96–100 %: without the $\langle$Trigger|Protocol Behavior|Fault Mode$\rangle$ triple, raw embedding matches thematically misaligned issues and the generator produces a VC for nearly every SYRS regardless of relevance. The full pipeline thus trades subset-level precision (58.0 % vs. 82.0 %) for suppression of this near-universal ungrounded generation—a precision/coverage choice we adopt to scale beyond the 50-item demonstration. We read this as a design trade-off, not a quality result: the volume the filter suppresses at corpus scale is not itself validated by human review—the proxy's acknowledged limit, valid only when filtering is active (§V-D).

c) IAA (FL condition).: On the 50-item FL evaluation, individual accept rates were E1 33/50 (66.0 %), E2 24/50 (48.0 %), E3 47/50 (94.0 %), E4 27/50 (54.0 %), for a majority vote (E2+E3+E4 $\geq$2/3) of 29/50 (58.0 %). Six rows produced E1–MV disputes—five where E1 accepted but MV rejected (ExtAddr scope mismatch or tool-layer contamination), one (Row 013) where E1 rejected but MV accepted (BufferedReader ECU wake-up boundary)—with MV prevailing per protocol. Mean 4-rater Gwet's AC1 is 0.63 for Abl-NF (substantial) and 0.37 for Abl-LM (moderate), corroborating the quality difference; pairwise agreement spans 66–92 % (Abl-NF) and 34–88 % (Abl-LM), the E3–E4 low driven by methodological differences on Extended-Addressing conditions (§VI).

D. RQ4: LLM Sensitivity

TABLE VI
RQ4: LLM SENSITIVITY. THE PRIMARY RUN IS HUMAN-EVALUATED (ACCEPT RATE); SENSITIVITY RUNS REPORT THE GENERATION-RATE PROXY ONLY (NOT COMPARABLE TO THE ACCEPT RATE). RUNS ARE INDEPENDENT RE-RUNS, SO QWEN-MAX'S CH-A RATE HERE (40.4 %) DIFFERS FROM THE MAIN RUN (41.8 %) BY NORMAL VARIANCE.

LLM	Ch-A	Ch-B
Primary run — human accept rate		
Qwen-Max	65.5 %	67.2 %
Sensitivity runs — generation-rate proxy		
Sentinel-adherent		
Qwen-Max (sensitivity)	40.4 %	93.7 %
Qwen-Plus	59.6 %	94.4 %
Non-sentinel-adherent (>93 % on both channels)		
DeepSeek-v4-Pro	99.0 %	100.0 %
DeepSeek-v4-Flash	100.0 %	99.3 %
Kimi-K2.6	94.2 %	97.2 %
GLM-5	98.6 %	98.6 %
Qwen3.6-Plus	100.0 %	99.3 %

a) Sentinel adherence separates two models from the other five.: Five of seven models generate for >93 % of items regardless of knowledge-source relevance, effectively ignoring the `NO_NOVEL_VC_FOUND` sentinel. Only Qwen-Max and Qwen-Plus emit it discriminatively, abstaining far more on Channel A's varied GitHub vocabulary than on Channel B (Table VI). Sentinel adherence is thus itself a model-selection signal: the proxy holds only on instruction-adherent models, where controlled abstention matters as much as raw accuracy. On both sentinel-adherent models (the two Qwen variants) the *generation-rate* B > A ordering holds; this is not the *accept-rate* ordering of RQ2, which, with only two adherent models, remains untested across families.

VI. Discussion and Threats to Validity

A. Failure Mode Analysis

Analysis of the 72 rejected VCs across Channels A and B reveals three failure modes that account for the channel acceptance gap. All three are residual gaps the Stage-5 expert review exists to catch: Field2VC proposes candidate missing VCs for an engineer to ratify, not a finished SYS.2 baseline.

FM-1: Boundary Underspecification (the most common mode). The VC is well-formed but lacks quantitative parameters (timing constraints, byte constants, thresholds), because source issues describe qualitative fault behaviour without numeric bounds and the LLM omits boundaries that appear only in the SYRS text. More common in Channel A; Channel B's component metadata and filtered fault modes constrain the generated bounds.

FM-2: Scenario Transplant (dominant Channel-B mode). The VC identifies the right ECU component and fault type but embeds the test in an irrelevant consumer-driving scenario lifted from an NHTSA narrative (e.g., "drive… 30 minutes with slight steering inputs"), making it impossible on a standard Hardware-in-the-Loop (HIL) bench.

FM-3: Filtering Collapse (thematic misalignment at scale). Without Stage 2 filtering, embedding similarity alone retrieves topically adjacent but thematically misaligned issues, and the LLM generates coherent VCs testing conditions unrelated to the SYRS under test (e.g., a UDS service-0x29 test for a framing requirement). This mode surfaces *at corpus scale*, not on the 50-item subset: across the full corpus Abl-NF generates for 96–100 % of items (§V-C), whereas the filtering-enabled pipeline abstains on the majority of Ch-A items—the gap is the volume of misaligned matches that filtering suppresses.

B. Threats to Validity

a) Construct validity.: The accept/reject label depends on raters applying the three-criterion protocol ($C \geq 3$ AND $N = \text{Y}$ AND $U = \text{Y}$). To mitigate subjectivity we recruited four raters—three independent scorers (E2–E4) and an adjudicator (E1)—with automotive RE backgrounds, and computed Gwet's AC1 [35] to account for the kappa paradox [37] under high Accept-class prevalence. E1 was not blind to the generation channel and is an author, a confirmation-bias risk—particularly for Channel B (67.2 %), which relied heavily on E1's adjudication. We anchored E1's rulings to the pre-registered protocol and cross-verified all 58 disputed Channel-B acceptances against the rulings of E2 (an independent ASPICE assessor). The Channel-A effect, notably, does not depend on adjudication on these items: restricted to the *agreed subset* (items both external raters labelled identically), Channel A is accepted on 37/37 items (100 %) versus 119/128 (93.0 %) for the baseline, while Channel B accepts only 51.3 %—so the Channel-A gain persists on clear-cut cases without adjudication, whereas the NHTSA gain leans more heavily on it. As a voluntary stress test we also report the headline rates under a conservative *no-E1 floor* (every split counted a reject; E1 resolved 50/87 Channel-A, 89/128 Channel-B, and 146/274 baseline splits): Channel A falls to 42.5 % against a 43.4 % baseline and Channel B to 15.6 %, so the per-channel uplift does not survive this worst case. Single-adjudicator, non-blind scoring thus remains a genuine limitation: unlike CrUISE-AC [4], which accepts on ≥3-of-4 *independent* experts, our protocol pairs two external raters with an adjudicator, and a larger blinded, independent panel under majority vote is the natural strengthening. The two lowest Completeness pairs both involve E4, reflecting disciplinary divergence between E4's executability perspective and E2/E3's requirements/process view, not measurement failure. The generation-rate proxy (RQ4) approximates novelty without human evaluation but cannot rank novel VCs by quality; its validity is confirmed only when filtering is active (§V-C).

b) Evaluator methodology divergence (Abl-LM).: The Abl-LM condition's low E3–E4 agreement (34 %, $\text{AC1} = -0.24$) is a systematic methodological difference, not noise: E3 judged Extended-Addressing preconditions on protocol-level merit (ISO 15765-2), whereas E2 and E4 treated them as deployment-blocking. Both are legitimate; majority vote lets the conservative position prevail, making the 58.0 % Abl-LM rate a lower bound.

c) Internal validity.: Stage 1 (Issue Collection) uses repositories chosen for ECU-adjacent vocabulary; we mitigate by drawing from three distinct stacks (UDS diagnostics, CAN transport, ECU toolchain). A small prompt-budget asymmetry remains: the do-not-repeat golden-VC context is truncated at 800 characters in the pipeline conditions but 600 in the baseline; we disclose rather than re-run, as it affects only that context, not the failure evidence. The absence of a hard similarity gate (§III-C) is supported post-hoc: every VC surviving sentinel abstention had a top-1 similarity ≥0.50, and above this floor similarity does not predict acceptance (quartile accept rates 57–76 %, non-monotonic); a hypothetical 0.60 gate would have discarded VCs with a *higher* accept rate (70.0 %) than those retained (63.5 %). The Channel-B trace (Table I) illustrates why: even when a retrieved complaint diverges in mechanism from the requirement, requirement-grounded generation—conditioned on the SYRS and its golden VC, not the evidence alone—keeps the VC anchored to SYRS intent, provided the requirement states that intent precisely enough.

d) Training-data exposure.: Public Channel-A issues are plausibly in the generator's pretraining data, a memorisation concern. The no-context baseline rules it out: the same Qwen-Max without retrieved evidence accepts only 45.3 %, so pre-training exposure is not applied unprompted—the in-context evidence drives the gain, an effect that holds on Channel B as well.

e) Evidence grounding vs. abstention.: One might ask whether the gains reflect the failure evidence or merely the Stage-2 abstention reporting only a favourable subset. The per-item yield (§V-A), which charges every abstention as a miss, separates the two: Channel B attains its $+21.3$ pp uplift while abstaining on only $9.2\,\%$ of items, so it cannot be selective silence, whereas Channel A abstains on $58.2\,\%$ and its yield (0.27) falls below baseline (0.35), trading recall for precision. We therefore claim failure grounding improves VC quality where filtered evidence aligns with requirement vocabulary, not that it uniformly raises generation; an abstention-matched head-to-head is left to future work.

f) DUT-boundary filtering failure (RQ3).: In the 50-item FL evaluation, 14 of the 21 MV-Reject VCs (67 %) arose from Device-Under-Test (DUT) boundary leakage: host-layer properties of the test libraries (log levels, file-descriptor limits, periodic-sender timing) were extracted as ECU-observable because Stage 2 cannot infer which side of the DUT boundary a property belongs to. For SYRS.0693_A, the generator asserted correctness at file-descriptor counts beyond 1023—a `python-can` property, not ECU-observable—which all but one rater rejected, confirming the evaluation is not rubber-stamping plausible output. A namespace guard for known test tools before triple extraction would suppress most such false positives.

g) External validity.: The primary evaluation covers one proprietary project. To probe generalisability we applied Channel A (Qwen-Max) to a separate Domain-2 pilot of 85 Diagnostics SYRS items (DCM, DEM, Diagnostic Services), which produced 15 novel VCs (17.6 % generation-rate proxy). A three-rater evaluation (E2–E4) accepted **13 of 15** (86.7 %, Wilson 95 % CI 62–96 %), exceeding the primary Channel-A rate (65.5 %), with substantial Novelty agreement (AC1 = 0.69) and moderate Usefulness agreement (0.42, lowered by E4's stricter executability criterion). The overlapping CI gives only preliminary evidence that cross-domain transfer succeeds without domain-specific prompting. Further limits: the B > A ordering holds only on two Qwen-family models, so cross-family stability is untested; powertrain, ADAS, and gateway domains remain unexamined; NHTSA effectiveness may not transfer where no regulator-operated complaint database exists; and the RQ3 ablation subset is homogeneous (all 50 items Communication-Stack).

h) Conclusion validity.: We apply Holm–Bonferroni correction [33] across three Fisher's exact tests (family-wise $\alpha = 0.05$), report Wilson 95 % CIs [31] for all rates, and report Cohen's h [34] alongside p-values to curb over-interpretation in small-to-medium samples. Fisher's exact test assumes independent observations; because VCs are nested within requirements and scored by a shared rater pool, that assumption holds only approximately, so we read the effect sizes and the agreed-subset analysis as robustness checks against the resulting clustering. Several scope-stratified cells (Table IV) are small (Diagnostics DEM and Diagnostic Services each n=6), with wide Wilson intervals; we report them as descriptive breakdowns, not powered comparisons.

VII. Conclusion

We presented Field2VC, a five-stage pipeline for completing automotive ECU verification criteria from empirical failure evidence. It grounds LLM generation in two complementary external channels operating at distinct abstraction levels—developer-reported GitHub defects (code level) and regulator-collected NHTSA complaints (vehicle level)—filtering each report into a structured fault representation and abstaining when no retrieved evidence aligns. Across a four-expert evaluation (three independent scorers and one author-adjudicator) of 489 generated VCs spanning 307 unique ASPICE SYS.2 requirements, failure grounding raised expert acceptance from a 45.3 % no-context baseline to 65.5 %–67.2 % (Holm-corrected $p < 0.001$, $h = 0.41$–0.45). Corpus-scale experiments further showed that abstention materially affected precision: without it, ungrounded generation became nearly universal (96–100 %).

These findings are best interpreted as characterising when and where failure grounding is effective rather than as a single headline acceptance rate. An adjudication-sensitivity analysis (§VI) showed the effect was uneven and partly adjudicator-dependent: within the adjudication-independent agreed subset (n=37), every GitHub-grounded VC was accepted, versus 93 % for the baseline, whereas the NHTSA channel leaned more heavily on adjudication. Beyond the headline rate, the results yield two transferable design rules. First, *a failure source is useful only where, after filtering, its vocabulary matches the quantitative, component-level language that safety-critical requirements demand*—regulator complaints satisfy this for body-control and chassis requirements, developer issues for the communication and diagnostics stack, and the two corpora are not interchangeable. Second, *the generation-rate proxy for novelty is valid only when the model honours an explicit generation-refusal instruction*—a precondition five of our seven models violated—so sentinel adherence must be verified before the proxy is trusted.

A Domain 2 pilot on 85 Diagnostics SYRS (UDS/DTC), where 13 of 15 novel VCs (86.7 %) were accepted without domain-specific prompting, gives preliminary evidence that the first rule transfers across subdomains. Collectively, these results answer the four research questions: failure grounding improves expert acceptance over a no-context baseline (RQ1); the gain depends on alignment between a requirement's sub-domain and its failure source's filtered vocabulary (RQ2); filtering and matching jointly drive quality, with abstention decisive at scale (RQ3); and the channel ordering holds across

LLMs only for models that honour the generation-refusal sentinel, which most do not (RQ4). To support replication, we release a reproducible benchmark (Data Availability): the 350-item SYRS corpus (307 evaluated) with expert-annotated golden VCs, all 489 generated VCs with per-rater labels and adjudication, the three-criterion protocol, the prompts, and the analysis scripts.

a) Future Work.: Three directions follow. The most pressing concerns evidential strength: because the headline rates rest partly on a single non-blind adjudicator (§VI), a larger blinded, independent panel under majority vote would convert the adjudication-dependent part of the NHTSA effect into a firm estimate. Second, formalising accepted VCs into machine-checkable predicates and linking each to its source failure record through a traceability matrix would turn one-shot generation into a baseline continuously re-grounded as new field evidence arrives, supporting the traceability expectations of ISO 26262. Third, we aim to operationalise the vocabulary-alignment rule into a predictive channel-selection model—predicting which channel and filter vocabulary yield acceptable criteria for a requirement subdomain—so that powertrain, ADAS, and gateway domains become instances of one model rather than a checklist of replications, and so the construction extends to other regulated domains (aviation, medical devices, rail) where specifications cannot anticipate deployment.

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